



Straight Lines: Gradients and Normals 1i

Subject & topics: Maths Stage & difficulty: A Level P1

Part A

Gradient of line

Find the gradient of the line l_1 which has equation $4x - 3y + 5 = 0$.

Part B

Perpendicular line

Find the equation of the line l_2 , which passes through the point $(1, 2)$ and is perpendicular to the line l_1 , giving your answer in the form $ax + by + c = 0$ where a, b and c are integers.

The following symbols may be useful: x, y

Part C

Midpoint

The line l_1 crosses the x -axis at P and the line l_2 crosses the y -axis at Q . Find the coordinates of the midpoint of PQ . State your answer using exact decimals.

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Part D

Length of PQ

Find the length of PQ .

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Logarithmic Plots 4

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

A student used a graph of $\ln y$ against x to discover that $y = e^{2x+5}$.

What were the gradient and intercept of the graph?

Part A

Find the gradient

What was the gradient of the graph?

Part B

Find the intercept

What was the intercept of the graph?

Adapted for Isaac Physics from NST IA Biology preparation work

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Simultaneous Equations 2i

Subject & topics: Maths Stage & difficulty: A Level P1

Solve the simultaneous equations

$$2x^2 + y^2 = 57$$

$$x + 2y - 6 = 0.$$

Enter the pair of x and y values that satisfy these equations that has the greatest value of x .

Part A

Value of x

Enter the value of x .

The following symbols may be useful: x

Part B

Value of y

Enter the value of y

The following symbols may be useful: y



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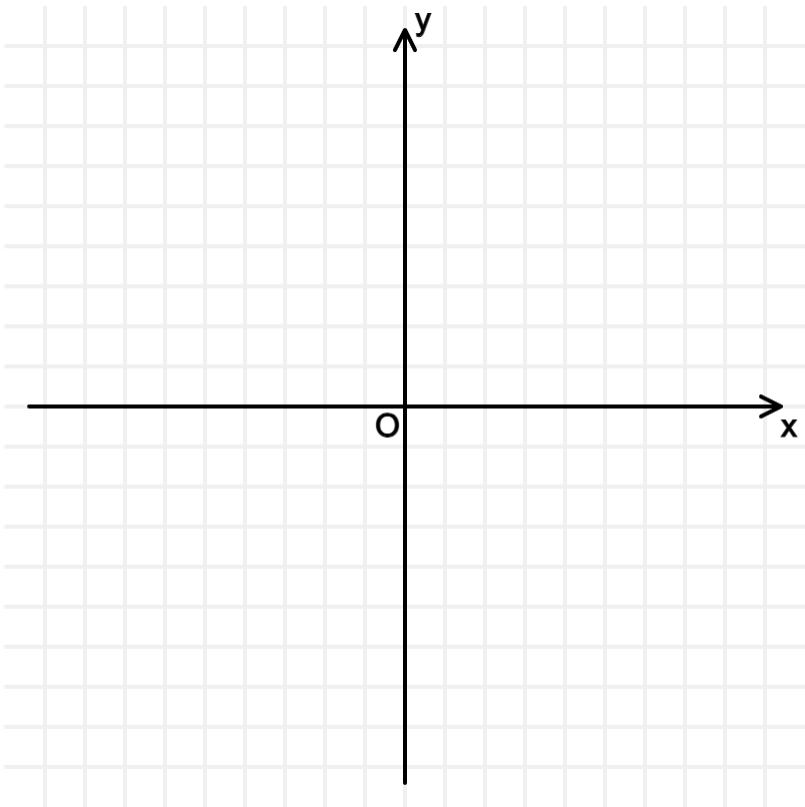
Quadratics: Graphs and Discriminants 1i

Subject & topics: Maths Stage & difficulty: A Level P1

Part A

Sketch graph

Sketch the curve $y = 2x^2 - x - 3$.



Part B

Solve inequality

Hence or otherwise solve the inequality $2x^2 - x - 3 < 0$.

Construct your answer from the items below.

Items:

<

>

x

$< x <$

$\leq x \leq$

$< x$ or $x <$

$\leq x$ or $x \leq$

$\leq$

$\geq$

-2

$-\frac{3}{2}$

-1

$-\frac{1}{2}$

0

$\frac{1}{2}$

1

$\frac{3}{2}$

2

Part C

Possible values

Given that the equation $2x^2 - x - 3 = k$ has no real roots, find the set of possible values of the constant k .

Write down an inequality for k .

The following symbols may be useful: $<$, $>$, k

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Completing the Square 1ii

Subject & topics: Maths Stage & difficulty: A Level P1

Part A

Complete square

Express $2x^2 + 12x + 13$ in the form $a(x + b)^2 + c$.

The following symbols may be useful: x

Part B

Solve equation

Solve the equation $2x^2 + 12x + 13 = 0$, giving your answers in the form $a \pm b$ where a and b are in simplified surd form.

The following symbols may be useful: $\pm$

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Quadratics: Functions of the Unknown 1i

Subject & topics: Maths **Stage & difficulty:** A Level P1

Find the roots of the equation $x - 8\sqrt{x} + 13 = 0$, giving your answers in the form $p \pm q\sqrt{r}$ where p , q , and r are integers.

The following symbols may be useful: $\pm$

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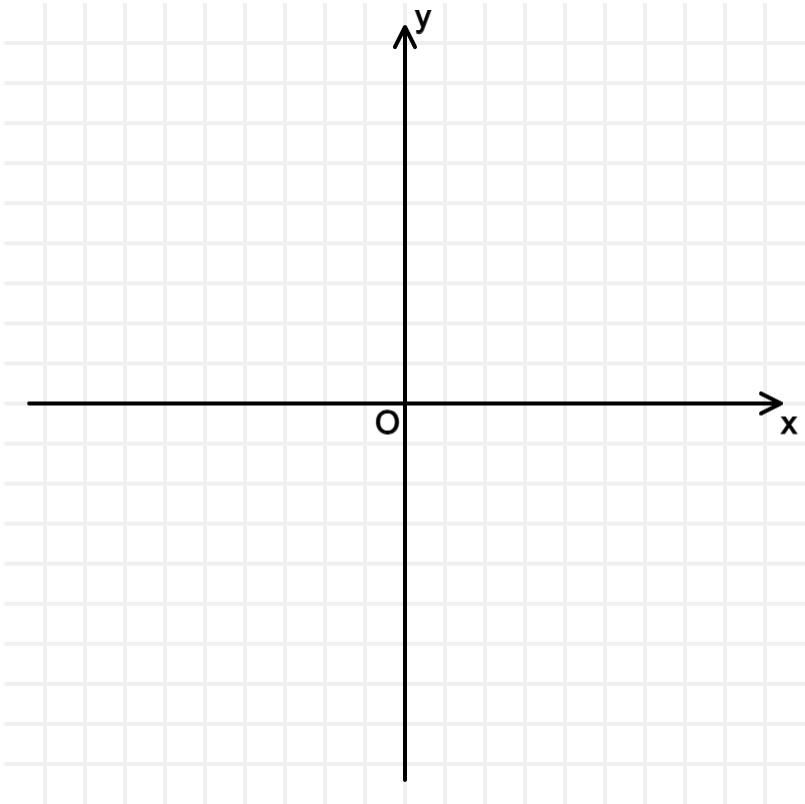
Exponentials and Logs

Subject & topics: Maths Stage & difficulty: A Level C2

Part A

Sketching

Sketch the curve $y = 6 \times 5^x$.



Part B

Find y -intercept

What is the value of the y -intercept of the curve $y = 6 \times 5^x$?

The following symbols may be useful: y

Part C

Find x -coordinate

The point P on the curve $y = 9^x$ has y -coordinate equal to 150. Use logarithms to find the x -coordinate of P.

Give the x -coordinate of P to 3 significant figures.

Part D

New x -coordinate

The curves $y = 6 \times 5^x$ and $y = 9^x$ intersect at the point Q.

Find the exact value of the x -coordinate at point Q, giving any logarithms in base three ($\log_3$).

When you are entering your answer, note that $\log_a b$ can be written using `log(b,a)`.

The following symbols may be useful: `log()`, `x`

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Solving Equations & Logs 2i

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Solve equation

Use logarithms to solve the equation $2^{n-3} = 18000$, giving your answer to 3 significant figures.

Part B

Simultaneous equations

Solve the simultaneous equations $\log_2 x + \log_2 y = 8$ and $\log_2\left(\frac{x^2}{y}\right) = 7$
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